## Amino acids

| | |
|---|---|
| PEPTIDE<br>PEPTJDE<br>PE(?PTI)DE | Amino acids are written as the standard IUPAC symbols. With O/U added as pyrolysine, and selenocysteine. J/B/Z are ambiguous amino acids for I/L, N/D, and Q/E respectively. Lastly X is added as any unknown amino acid. X is defined to have a mass of zero. So 'X[+435]' can be used to indicate a mass gap of 435 Da. |

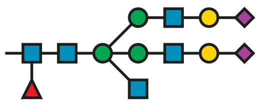
## Modification prefixes

| Mass | Name | Index | Description |
|---|---|---|---|
| U | U? | Unimod | Modifications from Unimod |
| M | M? | PSIMOD | Modifications from PSI MOD |
| R | R | Resid | Modifications from Resid |
| X | X | XLMOD | Modifications from XL-MOD |
| G | G | GNO | Modifications from GNOme |
| Obs | - | - | To indicate an observed mass |
| - | INFO | - | To add comments or description |
| - | Glycan | - | Glycan compositions |
| - | Formula | - | Molecular formulas |

## Modifications

Modifications are defined between square brackets '[mod]'. Multiple definitions can be combined with a vertical line '|' like so '[mod|mod|mod]'.

| | |
|---|---|
| Monoisotopic mass | PEPTI[+15.995]DE |
| Molecular formula | PEPTI[Formula:O]DE |
| Name (Unimod/PSI-MOD) | PEPTI[Oxidation]DE |
| Name specific ontology | PEPTI[U:Oxidation]DE |
| Index specific ontology | PEPTI[UNIMOD:35]DE |
| Comment or description | PEPTI[INFO:text]DE |
| Glycan composition | PEPN[Glycan:HexNAc5Hex5dHex1NeuAc2]ITDE |
| Glycan structure | PEPN[GNO:G75079FY]ITDE → Encodes: |
| Charged molecular formula | PEPTI[Formula:H-2Zn1:z+2]DE |



## Locations

**Localised**

| | |
|---|---|
| PEPT[mod][mod]IDE | (Multiple) modifications on the side chain of an amino acid |
| [mod][mod]-PEPTIDE | (Multiple) modifications on the N terminus |
| PEPTIDE-[mod][mod] | (Multiple) modifications on the C terminus |
| {mod}PEPTIDE | Labile modification, on this peptidoform |
| <mod@E,N-term>PEPTIDE | Fixed modification |

**Unlocalised**

| | |
|---|---|
| [mod]^n?PEPTIDE | On the entire peptidoform |
| PEPT[mod#name]ID[#name]E | On the given locations |
| PEP(TID)[mod]E | Within the given stretch |

**The order of all options when used together**

<mod@E>[mod]^n?{mod}[mod]-PEP-[mod]

**Using unlocalised rules**

PEP(TID[mod])[mod|Position:T,D|CoMKP]E

**Unlocalised additional rules**

| | |
|---|---|
| Position:E,N-term | Limit the allowed positions, uses the same positions as fixed modifications |
| Limit:2 | Limit the maximal number on one site, only for ^n global unlocalised modifications |
| CoMKP | Allows colocalising with placed modifications |
| CoMUP | Allows colocalising with other unlocalised modifications |

## Combining peptidoforms

| | |
|---|---|
| PEPTIDE+PEPTIDE | Chimeric peptidoforms (forming one compound peptidoform ion) |
| PEPT[mod#XLname]IDE//PEPT[#XLname]IDE | Cross-linked peptidoforms (forming one peptidoform ion) |
| PEPT[mod#BRANCH]IDE//PEP[#BRANCH]TIDE | Branched peptidoforms (forming one peptidoform ion) |
| PEPT[mod#XLname]IDE[#XLname] | Intra linked cross-linked peptidoform |

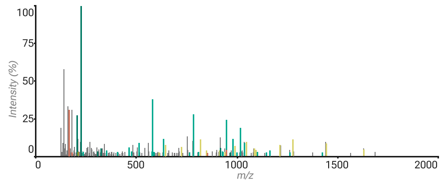
## Charge

| | |
|---|---|
| PEPTIDE/2 | Global charge, assumed to be protons |
| PEPTIDE/[Na:z+1,H:z+1] | Defined charge carriers, in this case 1 unlocalised sodium and 1 unlocalised proton |
| PEP[Formula:Na:z+1]TIDE/1 | Defined charge carriers, in this case 1 localised sodium and 1 unlocalised proton, this still has a global charge of 2 |

## Name

To add a description or metadata to an entire (compound) peptidoform (ion). To do this for single locations or stretches use the INFO tag.

(>Peptidoform name)PEPTIDE
(>>Peptidoform ion name)PEPT[mod#XLname]IDE//PEPT[#XLname]IDE
(>>>Compound peptidoform ion name)PEPTIDE+PEPTIDE




## Universal Spectrum Identifier

```
mzspec:<ID>:<file>:scan:<number>:<peptidoform>
                   index:<number>
                   nativeId:<number>(,<number>)*
```

mzspec:PXD023419:Peng2021_Herceptin_aLP.raw:scan:11368:RWGGDGFYAM[Oxidation]DYWGQGTLVTV/3

USIs are the URL for the mass spectrometry world. A USI points to a Proteomics Exchange Identifier (PXD...), a file, and a spectrum in that file. Additionally, a peptidoform can be added to the end, note that many servers require the peptidoform and a global charge state for the peptidoform.

ProForma v2.1
psidev.info/proforma
USI
psidev.info/usi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V1.0